

Week 4 Day 3 – Feature Engineering, Cross-Validation & Model Comparison

1. Engineered Features

Six new features were created using domain knowledge to improve model performance without introducing data leakage.

Feature	Reason
age_bucket	Represents different life stages.
hours_bucket	Captures different work patterns.
capital_gain_flag	Indicates investment income.
log_capital_gain	Reduces skewness of capital gain.
higher_education	Identifies college-educated individuals.
edu_hours	Combines education and working hours.

	Feature	Type	Creation Rule	Reason	Predictive Signal
0	age_bucket	Categorical	Age grouped into ranges	People of different age groups have different ...	Low (Grouped Mean)
1	hours_bucket	Categorical	Hours worked grouped into ranges	Working hours may influence yearly income.	Low (Grouped Mean)
2	capital_gain_flag	Boolean	capital_gain > 0	People with investments are more likely to ear...	MI = 0.0322
3	log_capital_gain	Numeric	log1p(capital_gain)	Reduces skewness of capital_gain while keeping...	MI = 0.0794
4	higher_education	Boolean	education_num >= 13	Higher education is often associated with high...	MI = 0.0490
5	edu_hours	Numeric	education_num × hours_per_week	Combines education and work effort into one fe...	MI = 0.0783

	Feature	Mutual Information
1	log_capital_gain	0.079405
3	edu_hours	0.078333
2	higher_education	0.048962
0	capital_gain_flag	0.032163

3. Cross-Validation Model Comparison

Three models were evaluated using identical preprocessing and **5-Fold Stratified Cross-Validation**.

- Logistic Regression
- Random Forest
- Gradient Boosting

The models were compared using **Precision (primary metric), ROC AUC, and F1 Score**. Logistic Regression produced the most consistent performance, while Random Forest and Gradient Boosting also achieved competitive results.

	Model	Precision (Mean ± Std)	ROC AUC (Mean ± Std)	F1 (Mean ± Std)
0	Logistic Regression	0.742 ± 0.010	0.912 ± 0.004	0.667 ± 0.005
1	Random Forest	0.720 ± 0.009	0.901 ± 0.003	0.662 ± 0.005
2	Gradient Boosting	0.780 ± 0.011	0.920 ± 0.003	0.680 ± 0.006

4. Statistical Comparison & Feature Importance

A paired **t-test** was performed between the two best-performing models to determine whether the performance difference was statistically significant. Feature coefficients from Logistic Regression and feature importances from Random Forest showed that **log_capital_gain**, **higher_education**, **capital_gain_flag**, and **edu_hours** were among the most influential engineered features, which aligns with business expectations.

Paired t-test
t-statistic : 8.7076
p-value : 0.001

Interpretation
t-statistic : 8.7076
p-value : 0.0010
The performance difference between Logistic Regression and Random Forest is statistically significant. This means the improvement is unlikely to be caused by random variation across folds.

5. Feature Selection & Final Recommendation

Feature selection using **SelectKBest** reduced training time but slightly decreased Precision, ROC AUC, and F1 Score. Therefore, the original and engineered features will be retained for Day 4 hyperparameter tuning.

Features Selected for Hyperparameter Tuning

- **Original Numeric:** age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week
- **Original Categorical:** workclass, education, marital-status, occupation, relationship, race, sex, native-country
- **Engineered:** age_bucket, hours_bucket, capital_gain_flag, log_capital_gain, higher_education, edu_hours

Feature Category	Features
Original Numeric	age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week
Original Categorical	workclass, education, marital-status, occupation, relationship, race, sex, native-country
Engineered	age_bucket, hours_bucket, capital_gain_flag, log_capital_gain, higher_education, edu_hours